2.4 Tree packing and arboricity

In this section we consider packing and covering in terms of edges rather than vertices. How many edge-disjoint spanning trees can we find in a given connected graph? And how few trees, not necessarily edge-disjoint, suffice to cover all its edges? These two questions have two classical theorems answering them. But rather than proving these theorems directly, we shall obtain them both as corollaries of a beautiful recent unification due to Bowler and Carmesin: the *packing-covering* theorem.

To motivate the tree packing problem, assume for a moment that our graph represents a communication network, and that for every choice of two vertices we want to be able to find k edge-disjoint paths between them. Menger's theorem (3.3.6) in the next chapter will tell us that such paths exist as soon as our graph is k-edge-connected, which is clearly also necessary. This is a good theorem, but it does not tell us how to find those paths; in particular, having found them for one pair of endvertices we are not necessarily better placed to find them for another pair. If our graph has k edge-disjoint spanning trees, however, there will always be k canonical such paths, one in each tree. Once we have stored those trees in our computer, we shall always be able to find the k paths quickly, between any given pair of vertices.

When does a graph G have k edge-disjoint spanning trees? If it does, it clearly must be k-edge-connected. The converse, however, is easily seen to be false (try $k = 2$); indeed it is not even clear that any edge-connectivity will imply the existence of k edge-disjoint spanning trees. (But see Corollary 2.4.2 below.)

Here is another necessary condition. If G has k edge-disjoint spanning trees, then with respect to any partition of $V(G)$ into r sets, every spanning tree of G has at least $r-1$ *cross-edges*, edges whose ends lie in different partition sets. (Why?) Thus if G has k edge-disjoint spanning trees, it has at least $k\,(r-1)$ cross-edges. This condition is also sufficient:

cross-edges

tree packing theorem

[8.6.9]

Theorem 2.4.1. (Nash-Williams 1961; Tutte 1961)
A multigraph contains k edge-disjoint spanning trees if and only if for every partition P of its vertex set it has at least $k\,(|P|-1)$ cross-edges.

Theorem 2.4.1 has a striking corollary: $2k$-edge-connectedness is enough to ensure the existence of k edge-disjoint spanning trees.

[6.4.4]

Corollary 2.4.2. *Every $2k$-edge-connected multigraph G has k edge-disjoint spanning trees.*

Proof. Every class in a vertex partition of G is joined to other partition classes by at least $2k$ edges. Hence, for any partition into r sets, G has at least $\frac{1}{2}\sum_{i=1}^{r} 2k = kr$ cross-edges. The assertion thus follows from Theorem 2.4.1. □

Note that the quantitative condition on cross-edges in Theorem 2.4.1 is equivalent to asking the same only for partitions into connected vertex sets: any other partition is refined by such a partition, and if the latter has enough cross-edges (even though it has more classes) then clearly so does the former. The tree packing theorem thus says that a multigraph has k edge-disjoint spanning trees as soon as all its contraction minors have enough edges to support k edge-disjoint spanning trees.

We shall meet Theorem 2.4.1 again in Chapter 8.6, where we prove an infinite analouge. This is based not on ordinary spanning trees (for which the result is false) but on 'topological spanning trees': the analogous structures in a topological space formed by the graph together with its ends, points at infinity that make it compact.

Let us now turn to the covering problem. To bring out its duality to the packing problem, we begin by rephrasing the latter. Let us say that some given subgraphs of a multigraph G form an *edge-decomposition* of G if their edge sets partition $E(G)$. Our spanning tree problem can now be recast as follows: into how *many connected* spanning subgraphs can we edge-decompose G? Since a spanning subgraph is connected if and only if it has an edge in every bond, the packing problem in this new guise has a 'dual' reminiscent of Theorems 1.5.1 and 1.9.4: into how *few acyclic* subgraphs – those whose complement meets all their circuits – can we edge-decompose G?

Let us say that some given graphs, not necessarily subgraphs of G, *cover its edges* if every edge of G lies in at least one of them. Our dual problem, then, is for which multigraphs G can we cover their edges by at most k trees. *cover*

An obvious necessary condition is that every set $U \subseteq V(G)$ induces at most $k\,(|U|-1)$ edges, no more than $|U|-1$ for each tree. Or, to phrase it dually to the tree packing condition, that no 'deletion minor' (subgraph) of G has too many edges to be covered by k trees.

Once more, this condition turns out to be sufficient too:

Theorem 2.4.3. (Nash-Williams 1964) *tree covering theorem*
The edges of a multigraph $G = (V, E)$ can be covered by at most k trees if and only if $\|G[U]\| \leqslant k\,(|U|-1)$ for every non-empty set $U \subseteq V$.

The least number of trees that can cover the edges of a graph is its *arboricity*. By Theorem 2.4.3, the arboricity of a graph is a measure for its maximum local density: it has small arboricity if and only if it is 'nowhere dense' in the sense that it has no subgraph H with $\varepsilon(H)$ large. *arboricity*

We finally come to the packing-covering theorem. Recall from Chapter 1.10 that when we form a contraction minor G/P of a multigraph G, we keep all the edges of G between different partition classes: edges between the same two classes $U, U' \in P$ become parallel edges of G/P.

packing-covering theorem

Theorem 2.4.4. (Bowler & Carmesin 2015)
For every connected multigraph $G = (V, E)$ and every $k \in \mathbb{N}$ there is a partition P of V such that every $G[U]$ with $U \in P$ has k edge-disjoint spanning trees and the edges of G/P can be covered by k spanning trees.

Before we prove the packing-covering theorem, let us deduce Theorems 2.4.1 and 2.4.3.

Proof of Theorem 2.4.1 (assuming Theorem 2.4.4).
Suppose a multigraph G has at least $k\,(|P|-1)$ cross-edges for every partition P of $V(G)$. Let P be the partition provided by Theorem 2.4.4. By the theorem, G/P has k spanning trees covering its edges. Since $\|G/P\| \geqslant k\,(|P|-1)$, they must be edge-disjoint. Combining them with the edge-disjoint spanning trees in the $G[U]$ that are also provided by Theorem 2.4.4, we obtain the desired k spanning trees of G. □

Proof of Theorem 2.4.3 (assuming Theorem 2.4.4).
Suppose every $U \subseteq V$ induces at most $k\,(|U|-1)$ edges in G. Let C be a component of G, and P the partition of $V(C)$ provided by Theorem 2.4.4. For each $U \in P$, each of the k edge-disjoint spanning trees of $G[U]$ that the theorem provides has $|U|-1$ edges, so all the edges of $G[U]$ lie in these trees. Combining these trees with the spanning trees of C/P that cover its edges, also provided by Theorem 2.4.4, we obtain k spanning trees of C covering its edges. These can be combined to k forests covering the edges of G. Add edges to turn these into the desired k trees. □

Given the power of the packing-covering theorem, its proof is strikingly short and elegant. To prepare some notation, consider a spanning tree T of G, a chord e, and an edge $f \in T$ on its fundamental cycle C_e. Then $T' = T + e - f$ is another spanning tree: this is immediate from Corollary 1.5.2, because T' is still connected and has the same number of edges as T. One says that T' is obtained from T by *exchanging* f for e.

$\mathcal{T}$, k

exchange chain

$i \mapsto j(i)$

Now let $\mathcal{T} = (T_j)_{j=1,\dots,k}$ be a family of spanning trees of G. Call a sequence $(e_0, \dots, e_n)$ of edges of G an *exchange chain* with respect to $\mathcal{T}$, *started by* e_0, if e_n lies on none of these trees but for every $i < n$ there exists $j =: j(i)$ such that e_{i+1} is a chord of T_j whose fundamental cycle with respect to T_j contains e_i.

$E(\mathcal{T})$

Let us write $E(\mathcal{T}) := \bigcup \{\, E(T) \mid T \in \mathcal{T} \}$ for any such family.

Lemma 2.4.5. *If e_0 starts an exchange chain with respect to $\mathcal{T}$ and lies in two of its trees, then there is a family $\mathcal{T}'$ of k spanning trees of G such that $E(\mathcal{T}) \subsetneq E(\mathcal{T}')$.*

Proof. Choose $(e_0, \dots, e_n)$ of minimum length among the exchange chains with respect to $\mathcal{T}$ that start with e_0. Then its elements e_i are distinct. And no e_i lies on the fundamental cycle with respect to any tree in $\mathcal{T}$ of any e_ℓ with $\ell > i+1$: if it did, we could omit $e_{i+1}, \dots, e_{\ell-1}$ from our sequence, contradicting its minimality.

For $i = 0, \dots, n$ inductively, we shall define families $\mathcal{T}^i = (T^i_j)_{j=1,\dots,k}$ of spanning trees of G such that, for all $i \leqslant \ell < n$, T^i_j

(i) $E(\mathcal{T}^i) = E(\mathcal{T})$;

(ii) e_i lies in two of the trees in the family $\mathcal{T}^i$;

(iii) $e_{\ell+1}$ is a chord of $T^i_{j(\ell)}$ and has the same fundamental cycle there as with respect to $T_{j(\ell)}$.

Note that these families have the same index set as $\mathcal{T}$, and we continue to use the map $i \mapsto j(i)$ from the indices of our exchange chain to the indices of $\mathcal{T}^i$ that we defined together with $\mathcal{T}$. $j(i)$

For $i = 0$ the family $\mathcal{T}^0 := \mathcal{T}$ is as required. Given $i < n$, let $\mathcal{T}^{i+1}$ be obtained from $\mathcal{T}^i$ by replacing $T^i_{j(i)}$ with $T^i_{j(i)} + e_{i+1} - e_i =: T^{i+1}_{j(i)}$ and letting $T^{i+1}_j := T^i_j$ for all other j. $\mathcal{T}^i$

By (iii) for i with $\ell = i$, the modified tree $T^{i+1}_{j(i)}$ is again a spanning tree of G: recall that e_i lies on the fundamental cycle of e_{i+1} with respect to $T_{j(i)}$, by definition of $j(i)$.

Assertion (i) for $i+1 < n$ follows from (i)–(iii) for i: we are not losing e_i by (ii), and we are not gaining e_{i+1}, since it lies on the fundamental cycle of e_{i+2} with respect to $T_{j(i+1)}$, and hence is already in $T^i_{j(i+1)} \in \mathcal{T}^i$ by (iii).

To verify (ii) for $i+1 < n$, note first that $j(i+1) \neq j(i)$, because e_{i+1} is a chord of $T_{j(i)}$ but lies on $T_{j(i+1)}$. Now $e_{i+1} \in T^i_{j(i+1)} = T^{i+1}_{j(i+1)}$ as noted above, as well as $e_{i+1} \in T^{i+1}_{j(i)}$, by definition of this tree.

For a proof of (iii) for $i+1 < n$ consider any $i+1 \leqslant \ell < n$. As long as $j(\ell) \neq j(i)$, we have $T^i_{j(\ell)} = T^{i+1}_{j(\ell)}$, and the assertion follows from (iii) for i. Now consider the case that $j(\ell) = j(i) =: j$. Then $e_{\ell+1}$ is a chord of T^i_j by (iii) for i, and $T^{i+1}_j = T^i_j + e_{i+1} - e_i$. As $e_{i+1} \neq e_{\ell+1}$ since $i \neq \ell$, our edge $e_{\ell+1}$ is still a chord of T^{i+1}_j. As noted at the start of the proof, its fundamental cycle with respect to T_j does not contain e_i. Hence (iii) for i implies (iii) for $i+1$.

It remains to check that $\mathcal{T}' := \mathcal{T}^n$ satisfies $E(\mathcal{T}) \subsetneq E(\mathcal{T}')$. As $e_n \in E(\mathcal{T}')$ by definition of $\mathcal{T}' = \mathcal{T}^n$, and $e_n \notin E(\mathcal{T})$ by assumption, all we have to check is that $E(\mathcal{T}) \subseteq E(\mathcal{T}')$. Now $E(\mathcal{T}) = E(\mathcal{T}^{n-1})$, by (i) for $i = n-1$. But the only edge we deleted from any tree in $\mathcal{T}^{n-1}$ when we turned it into $\mathcal{T}^n$ was e_{n-1}. By (ii) for $i = n-1$, this edge also lay on another tree in $\mathcal{T}^{n-1}$, which we left unchanged for $\mathcal{T}^n$. $\square$

Proof of Theorem 2.4.4. Let $\mathcal{T} = (T_1, \dots, T_k)$ be a family of k spanning trees of G, chosen with $E(\mathcal{T})$ maximal. Let D be the set of all edges of G that start an exchange chain with respect to $\mathcal{T}$. These include all edges not in $E(\mathcal{T})$, since they form singleton exchange chains. Let P be the partition of V into the vertex sets of the components of (V, D).

For the theorem's packing assertion, let $U \in P$ be given. For all $j = 1, \dots, k$ let S_j be the subgraph of T_j induced on U by its edges in D. These forests S_j are edge-disjoint, since by the maximality of $E(\mathcal{T})$ and Lemma 2.4.5 no edge in D lies in more than one T_j. Let us show that the S_j are connected.

Since the edges from D form a connected submultigraph on U, it suffices to show that for every edge $uu' \in D$ with $u, u' \in U$ there is a u–u' path in S_j. This is clear if uu' lies in T_j, and hence in S_j. If it does not, then the path uT_ju' still has all its edges e in D, and hence lies in S_j: if $e_0, \dots, e_n$ is an exchange chain witnessing that $e_0 = uu' \in D$, then $e, e_0, \dots e_n$ is an exchange chain putting e in D, because e lies on the fundamental cycle of e_0 with respect to T_j.

As every T_j induces connected subgraphs S_j on the partition classes of P, contracting these S_j turns the T_j into spanning trees T_j' of G/P. These T_j' cover all the edges of G/P, since $E \setminus E(\mathcal{T}) \subseteq D$. □

The packing-covering theorem differs from both the tree packing and the tree covering theorem in a fundamental way. The non-trivial directions of the latter two theorems each obtain a structural assertion about a graph, the existence of a packing or covering, as a consequence of quantitative assumptions about all their minors of a certain type: contraction minors for the packing theorem, and 'deletion minors' – i.e., subgraphs – for the covering theorem. This format makes them interesting: they offer valuable structural information for one graph in exchange for less valuable quantitative information about many smaller graphs.

The packing-covering theorem, by contrast, makes a structural assertion about every graph: with no need for any assumptions at all, neither quantitative nor qualitative.

The packing-covering theorem extends to infinite graphs in two interestingly different ways; see Exercises 20 and 131 in Chapter 8.

2.5 Path covers

Let us return once more to König's duality theorem for bipartite graphs, Theorem 2.1.1. If we orient every edge of G from A to B, the theorem tells us how many disjoint directed paths we need in order to cover all the vertices of G: every directed path has length 0 or 1, and clearly the number of paths in such a 'path cover' is smallest when it contains as many paths of length 1 as possible – in other words, when it contains a maximum-cardinality matching.

In this section we put the above question more generally: how many paths in a given directed graph will suffice to cover its entire vertex set? Of course, this could be asked just as well for undirected graphs. As it turns out, however, the result we shall prove is rather more trivial in